

Evaluating Language Models

Perplexity · BLEU · ROUGE · BERTScore · LLM-as-Judge · Benchmarks

Outline

The evaluation problem

Perplexity

n -gram overlap metrics

Model-based evaluation

Benchmarks & wrap-up

Why evaluation is hard

Prompt: "Summarize the key findings of the study on climate change."

Output A:

"The study found that global temperatures rose 1.2°C since 1900, with accelerating ice loss in the Arctic."

Output B:

"The research demonstrates significant atmospheric changes leading to enhanced precipitation patterns globally."

Multiple valid outputs

Many correct ways to say the same thing

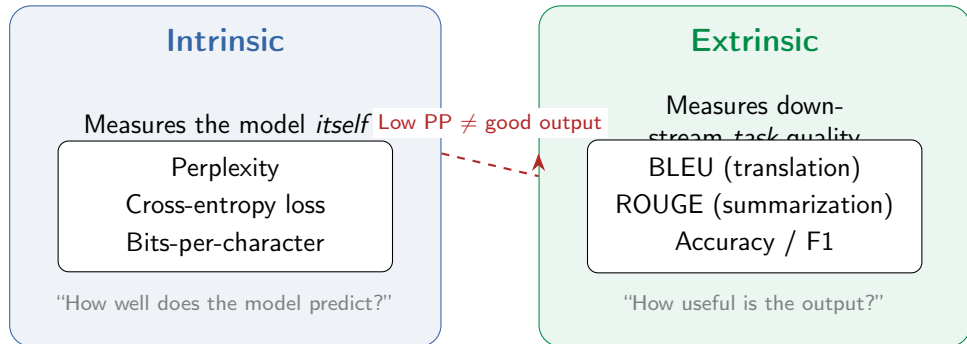
Meaning vs. form

Surface similarity $\neq$ semantic similarity

Task-dependent

Translation, summary, chat need different metrics

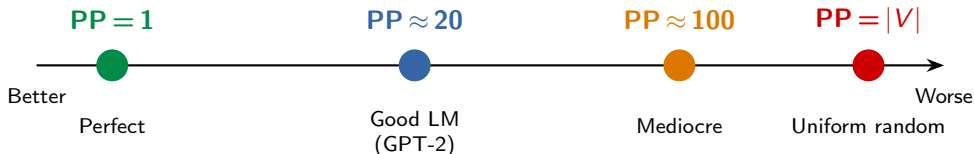
Intrinsic vs. extrinsic evaluation



Perplexity

$$\text{PP}(W) = \exp\left(-\frac{1}{N} \sum_{i=1}^N \log P(w_i \mid w_{<i})\right)$$

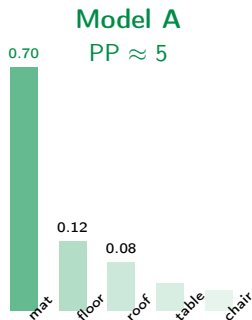
Intuition: On average, the model is as uncertain as choosing uniformly among **PP** options at each step.



Equivalently: $\text{PP} = \exp(H)$ where
 $H = -\frac{1}{N} \sum \log P(w_i \mid w_{<i})$

Perplexity — visual intuition

Predicting the next token after: “The cat sat on the _____”



Lower perplexity = model concentrates probability on the right tokens

Perplexity — caveats

1. Tokenizer-dependent

Different BPE vocabularies produce different N — PP scores across models with different tokenizers are *not* comparable.

2. Not a quality metric

Low PP means good prediction, not good *content*. A model can confidently produce fluent nonsense.

3. Memorization

A model that memorizes training data has very low PP on that data but fails on new text. Check on *held-out* data.

4. LM-only

Perplexity only applies to generative (autoregressive) models. For classification, QA, etc. use task-specific metrics.

Takeaway: PP is great for comparing LMs *on the same data and tokenizer*, but don't over-interpret it.

BLEU — Bilingual Evaluation Understudy

Papineni et al., 2002 — designed for machine translation

Reference:

The

cat

is

on

the

mat

match

no match

Candidate:

The

cat

sat

on

the

mat

$$p_n = \frac{\text{matched } n\text{-grams in candidate}}{\text{total } n\text{-grams in candidate}}$$

Unigram precision: $p_1 = 5/6 \approx 0.83$ Bi-

gram precision: $p_2 = 3/5 = 0.60$

Core idea: count how many n -grams in the candidate also appear in the reference

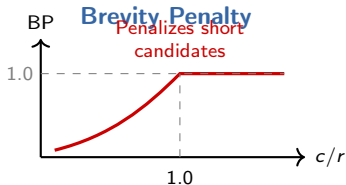
BLEU — modified precision & brevity penalty

Problem: “the the the the the” has 100% uni-gram precision against any sentence containing “the”!

$$\text{BLEU} = \underbrace{\text{BP}}_{\text{brevity penalty}} \cdot \exp\left(\sum_{n=1}^N w_n \log p_n\right)$$

$$\text{BP} = \min\left(1, \exp(1 - r/c)\right)$$

r = reference length, c = candidate length, $N = 4$, $w_n = \frac{1}{4}$



BLEU-4 is standard:
geometric mean of p_1, p_2, p_3, p_4
Higher $n \Rightarrow$ stricter fluency check

BLEU — worked example

Reference: “The cat is sitting on the mat”

$r = 7$

Candidate: “The the cat mat”

$c = 4$

Clipped precision by n -gram order

n	Candidate n -grams	Clipped matches	p_n
1	4	3	3/4
2	3	1	1/3
3	2	0	0/2
4	1	0	0/1

$$\text{BP} = \exp(1 - 7/4) = \exp(-0.75) \approx 0.47$$

$$\log\text{-avg} = \frac{1}{4}(\log 0.75 + \log 0.33 + \log \varepsilon + \log \varepsilon) \quad (\varepsilon = \text{smoothed zero})$$

BLEU $\approx$ **0.47** $\times$ (very small) $\approx$ **very low** — short, incomplete candidate

ROUGE — Recall-Oriented Understudy for Gisting Evaluation

BLEU asks: “How many candidate n -grams appear in the reference?” (**precision**)

ROUGE asks: “How many *reference* n -grams appear in the candidate?” (**recall**)

$$\text{ROUGE-}N = \frac{\text{matched } n\text{-grams}}{\text{total } n\text{-grams in reference}}$$

ROUGE-1

Unigram recall
Content coverage

ROUGE-2

Bigram recall
Fluency + content

ROUGE-L

Longest Common
Subsequence
Word-order aware

In practice, ROUGE-1, ROUGE-2, and ROUGE-L are all reported.
F1 variants (harmonic mean of precision & recall) are common.

BLEU vs. ROUGE

BLEU

Precision-based

“What fraction of candidate n -grams are correct?”

Best for: **Translation**

ROUGE

Recall-based

“What fraction of reference content is captured?”

Best for: **Summarization**

BLEU

ROUGE

Focus	Precision	Recall (+ F1)
Penalizes short output	Yes (BP)	No
Penalizes missing content	No	Yes
Typical n	1–4 (geometric mean)	1, 2, or L
Range	0–1 (often $\times 100$)	0–1

Limitations of n -gram metrics

Reference: “The movie was really excellent”

Candidate A:

“The film was truly great”

Good paraphrase, low BLEU

Candidate B:

“The movie was really bad”

Wrong meaning, high BLEU

Synonyms ignored

“great” $\neq$ “excellent” even though they mean the same

Semantics missed

“really bad” vs. “really excellent” share most n -grams

Multiple references needed

One reference can't capture all valid translations / summaries

Open-ended generation

No reference exists for creative writing, chat, etc.

Modern alternatives: **BERTScore** (embedding similarity), **METEOR** (synonyms + stemming), **COMET** (learned metric), **LLM-as-Judge**

BERTScore — embedding-level similarity

Zhang et al., ICLR 2020 — uses contextual embeddings instead of n -gram matching

Key idea: embed each token with a pre-trained model (e.g., RoBERTa), then compute similarity between reference and candidate tokens

Precision (P)

For each *candidate* token, find max-similarity reference token.
Average over

Recall (R)

For each *reference* token, find max-similarity candidate token.
Average over

F1

$$F_1 = 2 \frac{P \cdot R}{P + R}$$

Most commonly reported

Strengths:

- Captures synonyms & paraphrases
- Correlates better with human judgment
- Works across many languages
- No training needed (use off-the-shelf LM)

Limitations:

- Depends on choice of embedding model
- Slower than BLEU/ROUGE (GPU needed)
- Still reference-based: needs ground truth
- Sentence-level, not discourse-level

LLM-as-Judge — using LLMs to evaluate LLMs

Zheng et al., 2023 — now a dominant evaluation paradigm for open-ended generation

Core idea: use a strong LLM (e.g., GPT-4) as an automated judge to score or rank

MT-Bench

80 multi-turn questions
across 8 categories
Judge LLM scores 1–10
on helpfulness, relevance,
accuracy, creativity

Single model evaluation

Known biases:

- Position bias (prefers first response)
- Verbosity bias (longer = “better”)
- Self-enhancement (favors own outputs)
- Style over substance

Chatbot Arena (LMSYS)

Crowdsourced blind A/B tests:
humans pick preferred response

Elo ratings from 1M+ votes
(like chess rankings)

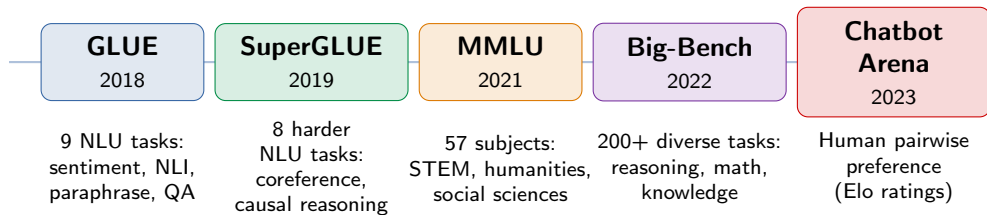
Gold standard for chat models

Why it dominates:

- Scales to any open-ended task
- No reference texts required
- High correlation with human judgment
- 10–100× cheaper than human eval

Benchmarks — evaluating holistically

Instead of one metric, aggregate many tasks to measure general capability



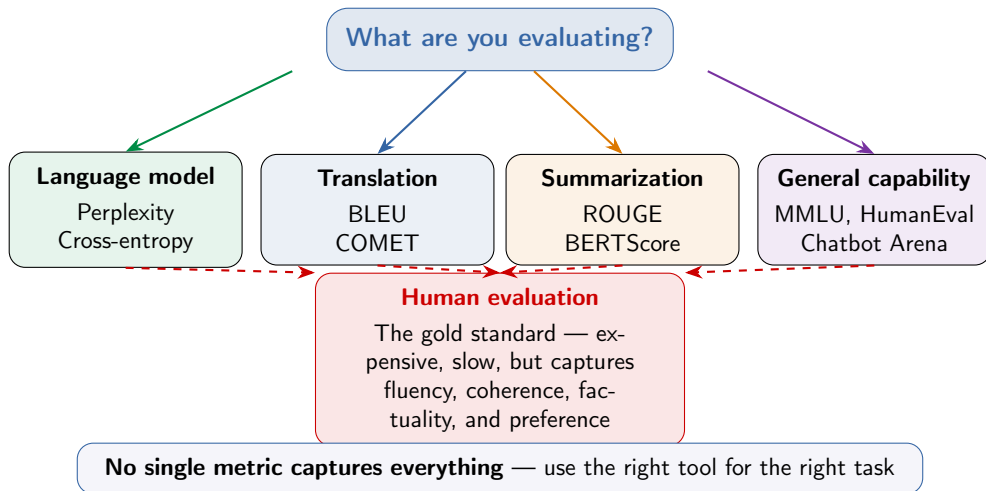
Key idea: single metrics are fragile — benchmarks aggregate many sub-tasks for a holistic view

Benchmark saturation: GLUE was “solved” within a year — SuperGLUE soon after. The bar keeps rising.

Key benchmarks at a glance

Benchmark	What it tests	Format	Notable for
MMLU	Knowledge (57 subjects)	Multiple choice	Breadth of knowledge
HumanEval	Code generation	Function completion	Coding ability
HellaSwag	Commonsense reasoning	Sentence completion	Physical understanding
TruthfulQA	Factual accuracy	QA	Hallucination resistance
GSM8K	Math reasoning	Word problems	Step-by-step reasoning
ARC	Science questions	Multiple choice	Grade-school science
Winogrande	Coreference	Pronoun resolution	Linguistic understanding
MMLU-Pro	Knowledge (harder)	Multiple choice (10-way)	Harder MMLU + reasoning
GPQA	Graduate-level science	Multiple choice	PhD-level physics/chem/bio
SWE-Bench	Coding agents	Real GitHub issues	End-to-end software eng.
LiveBench	Mixed (refreshed monthly)	Various	Contamination-resistant

The evaluation landscape



Further reading

Classic Metrics

- Papineni et al. (2002), “BLEU: A Method for Automatic Evaluation of Machine Translation”
- Lin (2004), “ROUGE: A Package for Automatic Evaluation of Summaries”
- Banerjee & Lavie (2005), “METEOR: An Automatic Metric for MT Evaluation”

Modern Evaluation

- Zhang et al. (2020), “BERTScore: Evaluating Text Generation with BERT”
- Zheng et al. (2024), “Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena”

Benchmarks & Leaderboards

- Srivastava et al. (2023), “Beyond the Imitation Game” (BIG-Bench)
- Hendrycks et al. (2021), “Measuring Massive Multitask Language Understanding” (MMLU)

Questions?

Next: Early Notable Models — GPT, BERT, T5